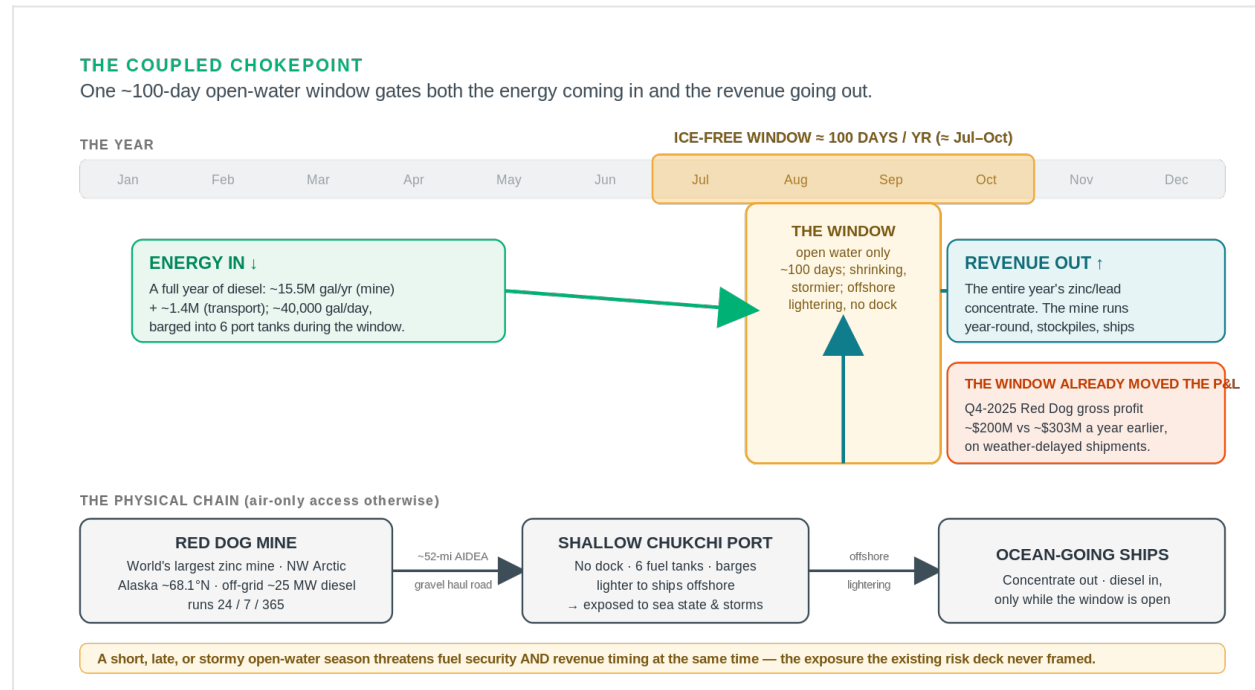


One Window In, One Window Out — on Ground That Won't Hold

One ~100-day window carries the whole year's fuel in and the whole year's revenue out — and the Arctic is closing it.

Red Dog Mine · Teck operates, NANA Regional Corp. owns the land · NW Arctic Alaska · Research-grounded (public sources + our methodology; not InfraSure MCP / model-gpr) · As of 2026-06-14, re-grounded 2026-06-15 · Internal draft



The coupled chokepoint — one ice-free window gates diesel-in and concentrate-out, and it has already moved the P&L (schematic). Sources: AIDEA; Teck FY2025 production + Q4-2025 results. As of 2026-06-15.

The claim

Red Dog is a **single-point-of-failure energy-and-logistics asset in the fastest-warming region on Earth**: one ~100-day ice-free window (≈ Jul–Oct) carries **both** the year's diesel **in** (~15.5M gal) and the entire concentrate export **out**, so a short, late, or stormy season threatens fuel security *and* revenue timing at once — already proven (Q4-2025 Red Dog gross profit ~\$200M vs ~\$303M a year earlier, on shipping delays). Layered on it: a permafrost water-balance crisis a US regulator has **formally conceded is unfixable** (ADEC's 10-year schedule to ~2036; "no reasonably available remedy to the thawing of permafrost"), pushing a ~\$279M adaptation capex onto a mine with ~6 years of ore left, an escalating royalty, and an owner mid-merger into a copper major. The tail risk — a diesel-tank or tailings-dam failure on thawing ground — is the **realized Norilsk-2020**

geometry (~\$2B).

Posture: a research-grounded, directional read — the decision-relevant numbers are undisclosed, and that is itself the finding. **Driver:** the Arctic warms ~4× the global average and models under-state it (Rantanen 2022).

1 · The coupled chokepoint — the framing the risk deck missed

The most decision-relevant climate linkage at Red Dog is not a single hazard, it is a coupling. One short open-water season carries the whole year in both directions, and the same Arctic trend is squeezing it from both ends:

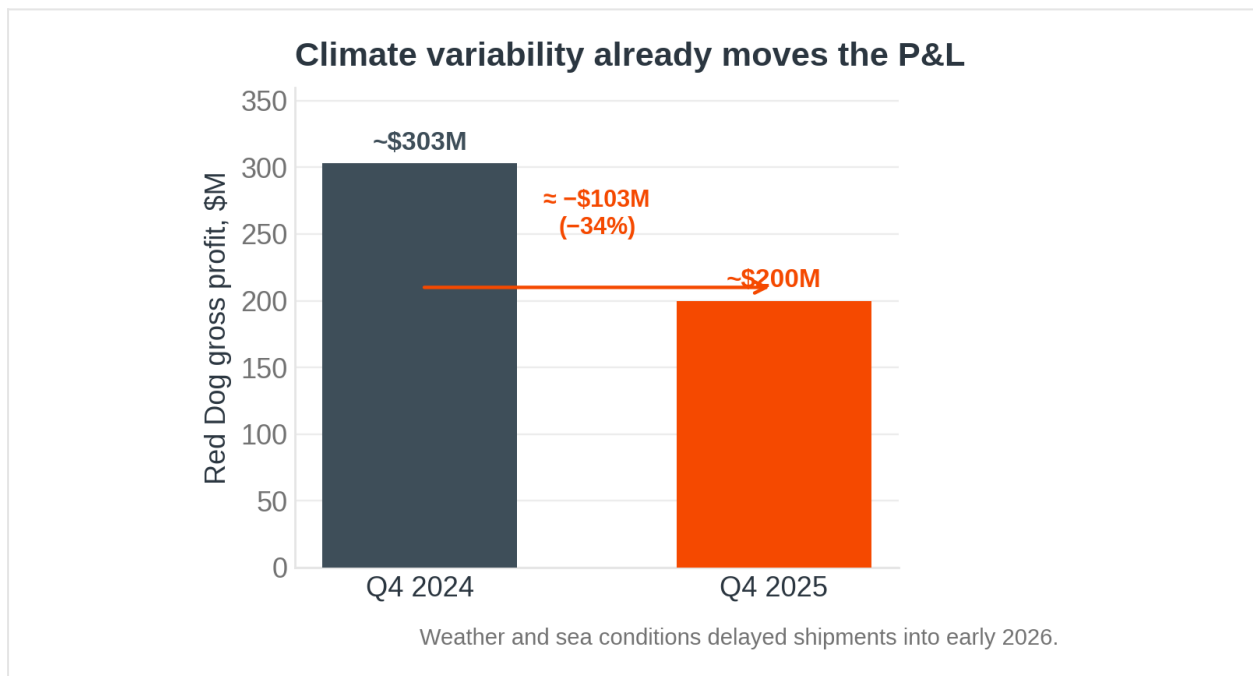
ONE ~100-day ice-free window (≈ July–October) carries BOTH:

Energy-in — the year's diesel: ~15.5M gal/yr (mine) + ~1.4M (transport); ~40,000 gal/day; barged into 6 port tanks.

Revenue-out — the entire annual concentrate export; the mine produces year-round, stockpiles, and ships only in the window.

→ a short, late, or stormy open-water season threatens fuel security AND revenue timing simultaneously.

This is not hypothetical. In late 2025, "*weather and sea conditions*" delayed shipments into early 2026, and **Q4-2025 Red Dog gross profit fell to ~\$200M (from ~\$303M a year earlier)** on the timing — climate variability already moving the P&L through the logistics window. (*Teck Q4-2025 results, Feb 2026 — fact/high.*)



Q4 Red Dog gross profit, 2024 vs 2025. Source: Teck Q4-2025 results (Feb 2026). As of 2026-06-15. fact/high.

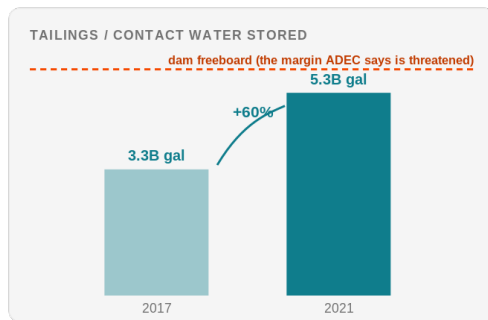
- **Sea-ice retreat cuts both ways:** it can lengthen the shipping window (operational upside) *but* removes the landfast-ice buffer that protected the coast — the same coastal cell is eroding fast at nearby Kivalina. The port relies on **offshore lightering**, so it is exposed to *sea state and storms*, not just ice. (*AIDEA; Ground Truth Alaska — fact/medium.*)
- **The single-window dependency is undisclosed at the point that matters:** how many days of fuel the site can run if a resupply season is missed or cut short is **not public** (see §5). An 8.8 MW solar array (Tugliq, ~2 MW summer-effective, tax-credit-timed to start mid-2026) dents diesel but does not change the resupply dependency. (*Alaska Beacon, Mar 2026 — fact/high; the "dents not displaces" read is ours.*)

2 · The ground won't hold — a water-balance crisis the regulator calls unfixable

The second pillar is documented in **primary regulatory text**, not press. A wetter Arctic on thawing permafrost has pushed the mine's water balance past what its infrastructure was built for, and the regulator has said so on the record:

THE GROUND WON'T HOLD

A wetter Arctic on thawing permafrost, and a regulator that says there is no fix.



ALREADY PAST THE DESIGN ENVELOPE

The Aug-2012 storm dropped **15.2 in of rain** and exceeded the tailings dam's **design flood** — the infrastructure has already operated outside what it was built for.

THE REGULATOR'S CONCESSION (ADEC, 2026-04-09)

"There is no reasonably available remedy to the thawing of permafrost and its impact on the receiving waters."

Permit reissued with a 10-yr compliance runway to ~2036 — outliving the ~2032 ore life.

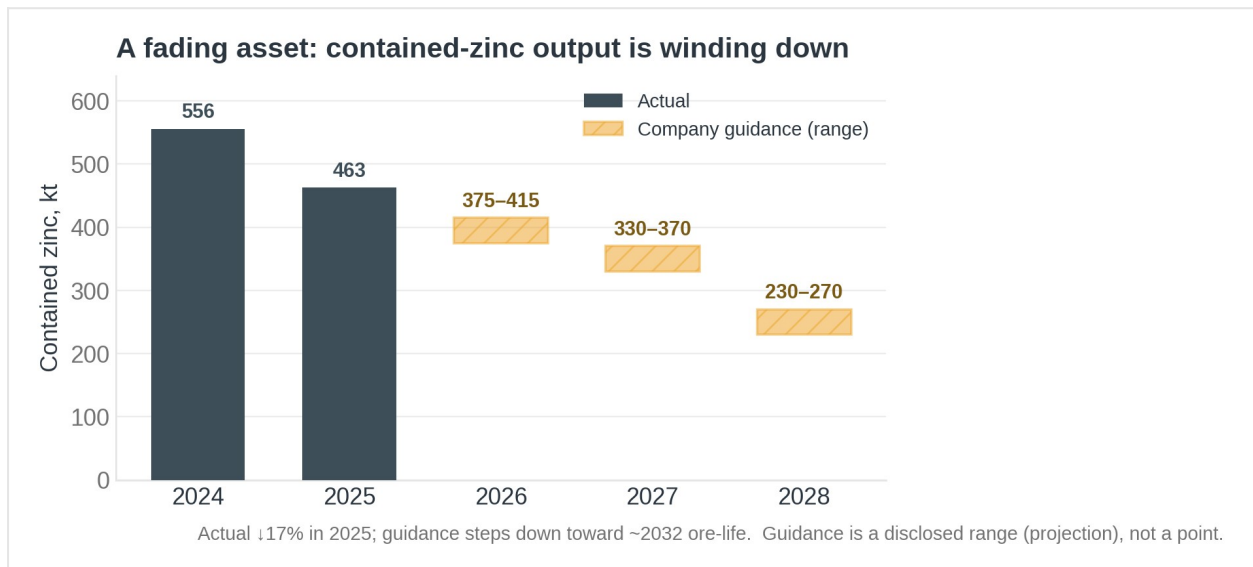
The ground won't hold — rising contact water against the dam-freeboard margin, a storm past the design flood, and the regulator's concession (schematic). Source: Alaska DEC APDES AK0038652 Statement of Basis (2026-04-09). As of 2026-06-15. fact/high.

- Alaska DEC is **reissuing the discharge permit (APDES AK0038652) with a 10-year compliance-schedule extension** (public-noticed; effective after May 19, 2026, to ~2036), stating in its own Statement of Basis (2026-04-09) that *"there is no reasonably available remedy to the thawing of permafrost and its impact on the receiving waters."* A US regulator formally conceding a Clean-Water-Act limit can't be met because of permafrost is **decision-relevant and unusual**. The runway now *outlives the ~2032 ore life*, shifting the binding constraint toward closure-phase water treatment. (ADEC SOB — fact/high.)
- ADEC states stored water *"threatens the safety of the tailings impoundment dams,"* with a tail risk of *"dam failure"* / untreated-drainage release. The **Aug-2012 storm (15.2 in rain) exceeded the tailings dam's design flood** — the infrastructure has already operated outside its design envelope. (ADEC SOB — fact/high; the single strongest data point that the asset is past-envelope.)
- The mitigation under evaluation is a **~\$279M, ~50-mile pipeline to the Chukchi Sea** — *"more than half of Red Dog's annual gross profit"* — one of four options, **not committed**. ~\$19M was already spent on thaw-driven wastewater in 2020 alone. (Northern Journal Dec 2025; Alaska Public Media 2020; ADEC SOB — fact/high; the pipeline is an estimate under evaluation — projection.)
- The regional driver is real and independent of the mine: **"river rusting"** — thaw-exposed sulfide weathering turning 75+ (200+) Brooks Range streams orange — with on-site seeps measured at **pH 3.2 and TDS to 11,300 mg/L**. (O'Donnell 2024 / NOAA Arctic Report Card 2025; ADEC SOB — fact/high.)

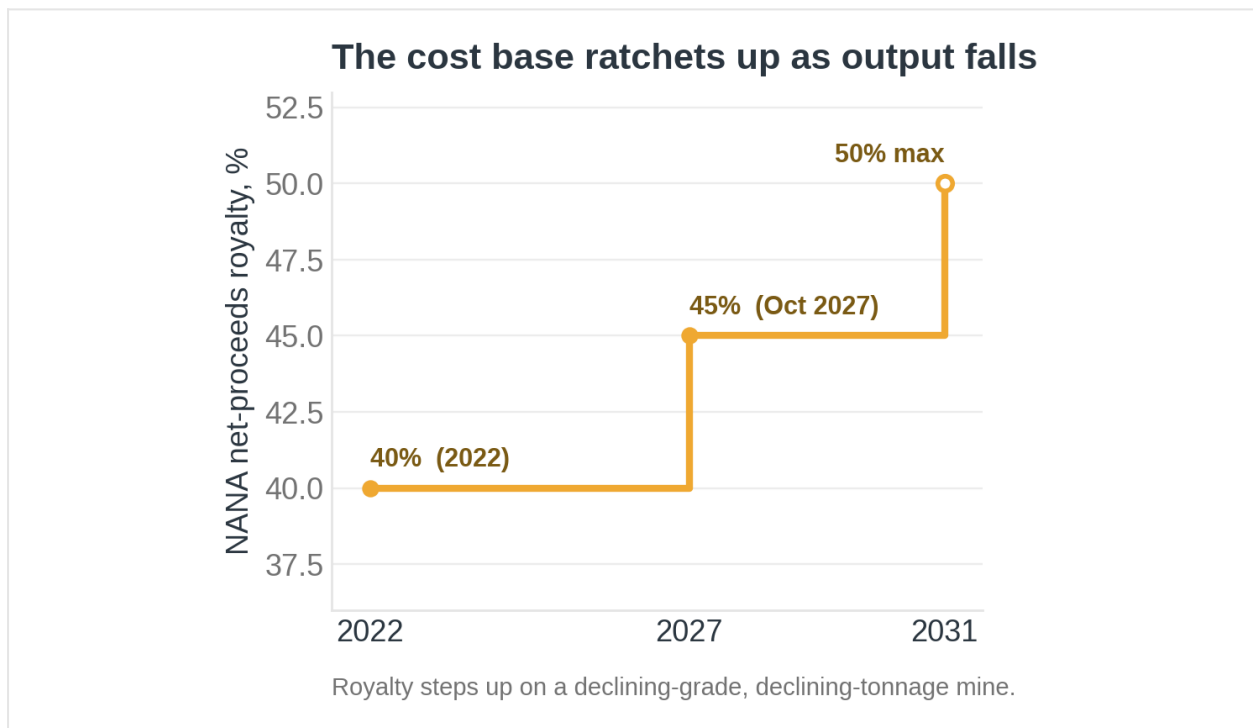
3 · The decision frame — a fading asset asked to fund a costly adaptation

What makes this a genuine decision, not a hazard list, is the collision of climate capex with late-life economics: output is falling, the cost base is rising, ownership is changing hands, and the liability outlives the revenue.

A fading asset	FY2025 462,700 t contained zinc (↓~17% from 555,600 t in 2024, on falling grades); ore ~2032; guided DOWN 2026 375–415 kt → 2027 330–370 → 2028 230–270 kt; extension (Aktigiruaq/Anarraaq) only pre-feasibility
A rising cost base	NANA royalty ratchets 40% (2022) → 45% (Oct 2027) → 50% max, as grades/tonnage fall
Ownership in flux	Anglo–Teck merger announced Sept 2025, shareholder-approved Dec 2025; Canada + S. Korea cleared, ONLY CHINA remains (as of Jun 2026), close ~Q3-2026–Q1-2027 → Red Dog a non-core zinc slice of a >70%-COPPER major
A long-tail liability	state closure assurance ~\$558M + PERPETUAL post-closure water treatment (outlives the revenue)



Contained-zinc output: actual (2024–25) then company guidance (2026–28). Sources: Teck FY2025 production (Feb 2026); production guidance + MD&A. As of 2026-06-15. Guidance is a disclosed range (projection).



NANA net-proceeds royalty schedule. Sources: Alaska DNR; North of 60 Mining News. As of 2026-06-15. fact/high (2031 step = 50% cap).

The sharp frame, from the Arctic-infrastructure literature: **near-term permafrost risk is largely mitigation-insensitive** — for an *existing* asset, **adaptation, not decarbonization, is the only lever** (Hjort et al. 2018, *Nature Comms*). So the question Red Dog's owner actually faces is **how much adaptation capex to sink into a depleting, soon-to-change-hands asset** — a ~\$279M pipeline (and "hundreds of millions over a decade" for a permanent fix) against ~6 years of declining ore. That tension — not a generic risk rating — is the read a decision-maker needs.

4 · The realized precedent — and the outlier class it opens

The hook is not a peer ranking (Red Dog is off-substrate, there is no fleet to rank); it is a **realized precedent**. The exact failure geometry has already happened, and it priced out at billions:

Norilsk, 2020. A diesel storage tank on thaw-weakened permafrost piles collapsed → **~21,000 t spill, ~\$2B in damages** — the exact Red Dog risk geometry, already realized. Norilsk already showed the bill. The one number that prices Red Dog's version — the **fuel-farm foundation engineering** — is the number that isn't public. (ERM / press — fact/high.)

Red Dog instantiates a class InfraSure will meet again: the **remote, off-grid, cold-region industrial asset** — *islanded on a diesel microgrid + dependent on a single climate-sensitive logistics window + built on degrading permafrost*. The method transfers to the next one (another mine, a port, a remote data center). The **mirror precedent** is the winter ice road (Tibbitt–Contwoyto, serving NWT diamond mines): the same narrowing-seasonal-window risk, *opposite sign* — they need ice *present*; Red Dog needs it *absent*. That symmetry is what makes "the single climate-sensitive window" a portable concept. (*Reuters* — *fact/high*.) Frameworks that anchor the class: **IPCC AR6 WG2 CCP6** ("warming and wetting"; most Arctic infrastructure impacted by mid-century, high confidence); **Hjort 2018** (33–69% of pan-Arctic infrastructure on thawing permafrost by 2050); **AMAP/SWIPA**.

5 · What we can't know — and why that's the finding

A disciplined read names its blind spots rather than guessing. For Red Dog, the **most decision-relevant numbers are not public** — and naming them precisely is exactly what a pilot or data-exchange would resolve:

Days of fuel autonomy	if an open-water window is missed/short — undisclosed (the latent single-point-of-failure)
Tailings-dam factor-of-safety	current freeboard margin — undisclosed (ADEC says it's threatened; the margin isn't public)
Fuel-farm foundation engineering	piles to bedrock? thermosyphons? — undisclosed (this is exactly the Norilsk failure mode)
Asset-level financials	Red Dog has no standalone public P&L (reported inside Teck's zinc segment; ~CAD \$846M of the segment's \$1,143M FY2025 gross-profit-before-D&A is the closest anchor) — so margin-at-risk can't be priced precisely

What's still resolving — and the forward door

The Arctic is warming ~4× the global average and faster than models assume, so the open-water window and the permafrost water balance keep moving; this is a directional read that firms as the Anglo–Teck merger, the production guidance, and the ADEC permit resolve. That uncertainty is the world's, not a gap in the work.

This is the rigor on public data. The per-asset quantified layer — fuel-autonomy modeling, tailings-dam factor-of-safety, the priced tail risk — is where InfraSure goes deeper; \$5

names exactly the undisclosed numbers a pilot or data-exchange would close. Talk to the team to take it there.

Sources & provenance

Primary / regulatory	Alaska DEC APDES AK0038652 Statement of Basis (2026-04-09) · Alaska DNR Large Mines (Red Dog) · AIDEA (DMTS)
Company	Teck Q4/FY2025 results (Feb 2026) · Teck production guidance + MD&A · Teck–Anglo merger releases (Sept/Dec 2025)
Peer-reviewed	Rantanen 2022 (Arctic ~4× warming) · Hjort 2018 (pan-Arctic infra exposure) · O'Donnell 2024 (river rusting) · IPCC AR6 WG2 CCP6
Reporting	Northern Journal / Alaska Beacon / Anchorage Daily News (permafrost, solar, fuel) · Fastmarkets · North of 60 Mining News · Ground Truth Alaska (critical-stakeholder)
Precedent	Norilsk 2020 diesel-tank collapse (ERM / press) · Tibbitt–Contwoyto winter ice road (Reuters)
Re-ground 2026-06-15	Teck 2025 Production & Sales Update (teck.com, 2026-02) · merger: Canada + S. Korea cleared, China pending (MLex; mining.com, 2026) · ADEC AK0038652 reissuance public notice + SOB (2026-04-09)

*This report renders the gated studio brief `studio/red\_dog\_mine/arctic\_climate\_risk.md`; full per-claim provenance is in the research record behind `docs/discussion/2026-06-14\_outlier\_method\_red\_dog.md`. **Research-grounded; not InfraSure MCP / model-gpr data.***